MAT1190 Lecture Notes

Arky!! :3c

’26 Fall Semester

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Day 1: Introduction (Sep. 8, 2026)

This is the first part of a two-semester course. Throughout this class, we will cover Hartshorne Chapter II. To quote Hunter Spink, "if I just went through Hartshorne straight, it would not make any sense; this textbook does not make any sense. As a first exposure, the book is not reasonable. " We will get through scheme theory, though. The goal of this class is to develop scheme theory as a main topic, then show how it applies to a bunch of easier theories. "A lot of people think schemes are so confusing, that you basically have to do 6 to 9 on a completely obsolete theory to show you how it actually matters" (but he won't do that).

Three references that are very useful, outside of Hartshorne.

(i) Ravi Vakil's *The Rising Sea*. This is meant to be a yearlong textbook.

(ii) Jo Harris' *The Geometry of Schemes*. Harris just says "this is schemes, suck it up" then develops a bunch of properties about it.

(iii) Qin Li's *Arithmetic* (something something).

The class will have an *oral* midterm and final (roughly October and December). There will be ungraded exercises (biweekly, as we go?) to turn in throughout the semester; the orals will be based on the easier questions.

This class will start with the theory of manifolds over $\mathbb{R}$, i.e., the classical theory of nice spaces. Consider the following insight on the Gelfand–Kolmogorov theorem: consider $C^0(M,\mathbb{R})$, the ring of continuous functions $M \to \mathbb{R}$ on a manifold M. How do we recover M from $C^0(M,\mathbb{R})$? Colloquially, different points of M see different aspects of any function in $C^0(M,\mathbb{R})$. If we start with a point $p \in M$, we get an $\mathbb{R}$-algebra map $C^0(M,\mathbb{R}) \to \mathbb{R}$ given by evaluating the functions at the given point p; we call this the "evaluation map", denoted

$$\mathrm{ev}_p \colon C^0(M,\mathbb{R}) \to \mathbb{R}, \quad f \mapsto f(p).$$

Specifically, $\mathrm{ev}_p \in \mathrm{Hom}_{\mathbb{R}-\mathrm{alg}}(C^0(M,\mathbb{R}),\mathbb{R})$. What does it mean for ev_p to be an $\mathbb{R}$-algebra map? As an example, while $\mathrm{Hom}(\mathbb{C},\mathbb{C}) = \mathrm{Aut}(\mathbb{C})$ contains "uncountably many horrible things", we see that $\mathrm{Hom}_{\mathbb{R}-\mathrm{alg}}(\mathbb{C},\mathbb{C})$ consists of homomorphisms that fix $\mathbb{R}$, i.e., it only admits the identity and complex conjugation.

Claim 1.1. $\{\mathrm{ev}_p\}_{p\in M}$ are the only elements of $\mathrm{Hom}_{\mathbb{R}-\mathrm{alg}}(C^0(M,\mathbb{R}),\mathbb{R})$.

We will do the proof for $M = \mathbb{R}$, then see that the proof follows analogously for several other examples of M as well.

Proof. Assume we have an $\mathbb{R}$-algebra map $\Phi\colon C^0(M,\mathbb{R}) \to \mathbb{R}$ that sends the identity function x to a, i.e., $x \mapsto a$. We claim that this map is ev_a; if not, then there is some function $g \in C^0(M,\mathbb{R})$ such that $g \not\mapsto g(a)$ under Φ. Indeed, consider $\ker(\Phi)$; we see that $x - a \in \mathrm{Ker}(\Phi)$; now, consider the function

$$(x-a)^2 + (g(x)-b)^2,$$

where $b := \Phi(g(x))$. This function was constructed to never vanish; the only way for it to be zero would be for $x = a$ and $g(x) = \Phi(g(x))$, but this never happens, so it is nonzero everywhere. However, $x - a \in \ker\Phi$ and $g(x) - b \in \ker\Phi$, so

$$(x-a)^2 + (g(x)-b)^2 \in \ker\phi,$$

whence it is a unit; this would mean

$$1 = \frac{1}{(x-a)^2 + (g(x)-b)^2} \cdot \Big((x-a)^2 + (g(x)-b)^2\Big) \in \ker\Phi,$$

which is contradictory. □

Following Grothendieck, let R be any ring, and consider Spec R, which is a set of points with a topology, and is a locally ringed space (i.e., local functions "glue" to form global functions). We will rigorously define Spec R later. The key insight about manifolds is that, since they consist of charts that are compatible to each other locally, it allows us to "glue" data from functions together. Another important thing to know about locally ringed spaces is that if a function is non-vanishing everywhere, then it is invertible. Together, this lets you recover a topology on your ring R. Note that this stuff is still intentionally hand-wavey for now.[1]

Consider the polynomials over the real numbers, $\mathbb{R}[x]$. It is a ring. What would happen if we tried to reconstruct the space according to the procedure described earlier? Let's look at the $\mathbb{R}$-algebra homomorphisms from $\mathbb{R}[x] \to \mathbb{R}$; then we get that it is $\mathbb{R}$, since $x \mapsto a \in \mathbb{R}$, and there is probably some way to proceed from here.

There is, however, a reason we don't do algebraic geometry over the real numbers; that reason being, the process would suggest that $\mathbb{R}[x]$ is that ring of functions on the real line. If we looked at $x^2 + 1 \in \mathbb{R}[x]$, we would see that it's a polynomial that doesn't vanish anywhere and not invertible. This makes these "space-recovery" arguments impossible, specifically the part where a "non-vanishing function is invertible". There are two solutions to this behavior... either you invert functions which don't vanish everywhere, so you map $\mathbb{R} \to \mathbb{R} \setminus \{0\}$, or you can add points for every possible evaluation to every field. For example, consider

$$\mathrm{ev}_i : \mathbb{R}[x] \to \mathbb{C}, \quad x \mapsto i;$$

we would see that $x^2 + 1 \mapsto 0$. Next time, we will define that Spec(R) is built out of all homomorphisms from $\mathbb{R}$ to fields, modded out by "some stuff".

[1] "you... uncover something so unbelievably terrible that i can't even begin to describe it. something that's going to ruin your life for the next... twelve weeks!"

Day 2: Spectrum of a Ring (Sep. 10, 2026)

Last class, we talked about evaluation maps. Let M be a manifold, and let $p \in M$. We have that all of the $\mathbb{R}$-algebra maps $C^0(M,\mathbb{R}) \to \mathbb{R}$ were given by $\mathrm{ev}_p : C^0(M,\mathbb{R}) \to \mathbb{R}$. A key fact about locally ringed spaces is that any function on them which is nonvanishing everywhere is invertible; when working with polynomials $\mathbb{R}[x]$, though, we see that x^2+1 constitutes a counterexample to the aforementioned statement.

The function x^2+1 maps $\mathbb{R}$ to $\mathbb{R}\setminus\{0\}$. Observe that we may identify $xy-1=0$ with $\mathbb{R}\setminus\{0\}$ by the isomorphism $(x,y)\mapsto x$; the curve $xy-1=0$ (visualized, perhaps, in the xy-plane) admits an automorphism $(x,y)\mapsto(y,x)$; if we pre-compose $(x,y)\mapsto x$ with this, we see that we get an automorphism on $\mathbb{R}\setminus\{0\}$ in the form of $x\mapsto x^{-1}$. This is called the *Rabinowitsch trick*.

Our solution to the above problem is to "add more points". Consider $\mathrm{Hom}_{\mathbb{R}-\mathrm{alg}}(\mathbb{R}[x],\mathbb{C})$ (note that we extended the codomain $\mathbb{R}$ to $\mathbb{C}$). Fix a field K, and consider $f_1,\dots,f_r \in K[x_1,\dots,x_n]$, and consider the set

$$\{f_1(x_1,\dots,x_n)=f_r(x_1,\dots,x_n)=0\}\subset K^n.$$

This is the zero set $z_K(f_1,\dots,f_r)$.

Theorem 2.1. We have the canonical bijection,

$$\mathrm{Hom}_{K-\mathrm{alg}}\left(\frac{K[x_1,\dots,x_n]}{(f_1,\dots,f_r)},K\right)\cong Z_K(f_1,\dots,f_r).$$

Proof. The key fact about polynomial rings is that, in general, for any ring R, $\mathrm{Hom}_{R-\mathrm{alg}}(R[x_1,\dots,x_n],S)$ is in a very natural bijection with n-tuples of elements of S. Concretely, you can send $(x_1,\dots,x_n)\mapsto(s_1,\dots,s_n)$ because $x_1,\dots,x_n$ don't have to satisfy any relation over R. In this manner, any map from the quotient ring to K (in the original theorem statement) must send $f_1,\dots,f_r$ to 0. Per our key fact, it follows that $f\mapsto \mathrm{ev}_{(s_1,\dots,s_n)}=f(s_1,\dots,s_n)$.

The canonical bijection thus arises for $(a_1,\dots,a_n)\in Z_K(f_1,\dots,f_r)$ and its associated evaluation maps $\mathrm{ev}_{(a_1,\dots,a_n)}$. □

Thus, we want to consider $K[x_1,\dots,x_n]/(f_1,\dots,f_r)$ to be the "ring of functions". We want to ask, what is the function f_1 on the space $Z_k(f_1,\dots,f_r)$? Clearly, $f_1|_{Z_k(f_1,\dots,f_r)}=0$ (and the same holds for each of $f_1,\dots,f_r$).

As an example, consider a field K, $f_1=x$, and $f_2=y$. Then $Z_K(f_1,f_2)=(0,0)$, and $K[x,y]/(f_1,f_2)\cong K$. If $f_1=x$ an d$f_2=y^2$, then

$$\frac{K[x,y]}{(x,y^2)}\cong\frac{K[y]}{y^2}.$$

Here, $\mathrm{ev}_{(0,0)}(y)=0$, but we aren't setting y to 0 (as we are quotienting out by y^2). Thus, we have a hurdle to deal with; (not all) functions will not be determined by their points.

Let L/K be a field over K, and consider $Z_L(f_1,\dots,f_r)$, where $f_1,\dots,f_r$ are polynomials over K. Since K is embedded in L, we can consider $f_1,\dots,f_r$ as polynomials over L as well, and write

$$Z_L(f_1,\dots,f_r)=\{(a_1,\dots,a_n)\mid f_1(a_1,\dots,a_n)=\cdots=f_r(a_1,\dots,a_n)=0\}\subset L^n.$$

He then drew a diagram.

Theorem 2.2. (i) $\mathrm{Hom}_{K-\mathrm{alg}}(K[x_1,\dots,x_n]/(f_1,\dots,f_r), L)$ is in bijection with $Z_L(f_1,\dots,f_r)$.

(ii) Let $R = K[x_1,\dots,x_n]/(f_1,\dots,f_r)$. We have that

$$\mathrm{Hom}_{K-\mathrm{alg}}(R, L) \cong \{(\text{prime ideals } p \subset R), \kappa(p) \xhookrightarrow{K-\mathrm{alg}} L)\},$$

where $\kappa(p) = \mathrm{Frac}(R/p)$.

Basically, this theorem says that "instead of being really good at solving system of equations, you could be really good at finding prime ideals instead."

Proof. For (i), the proof is the same as earlier. For (ii), we have a K-algebra map $\Phi\colon R \to L$; because the image is a field, we can factor the map through intermediate rings canonically. Because $\mathrm{Im}\,\Phi$ is an integral domain, Φ factors through

$$R \to R/p \xhookrightarrow{K-\mathrm{alg}} L,$$

where $\mathrm{Im}\,\Phi = R/p$. In particular, since L is a field, we can further write

$$R \to R/p \hookrightarrow \mathrm{Frac}(R/p) \xhookrightarrow{K-\mathrm{alg}} L.$$

The first two maps are canonical to the prime ideal p. □

"To solve a system of equations over a larger field, we specify a prime ideal, then specify an embedding."

Definition 2.3. We define $|\mathrm{Spec}(R)| \coloneqq \{p \subset R \mid p \text{ prime}\}$.

Corollary 2.4. If R is a field, then $|\mathrm{Spec}(R)|$ is a single point.

Of all the points that constitute $\mathrm{Spec}(R)$, of which they are prime ideals, some of them are $\ker \mathrm{ev}_p$ for $p \in Z_K(f_1,\dots,f_r)$.

As an example, consider $|\mathrm{Spec}\,\mathbb{R}[x]|$. $\mathbb{R}[x]$ is a UFD, so its prime ideals are of the form (f) with f irreducible. Specifically, $|\mathrm{Spec}\,\mathbb{R}[x]| = \{(x-a) \mid a \in \mathbb{R}\} \cup \{(x-a)^2 + b^2 \mid a \in \mathbb{R}, b \sim -b \in \mathbb{R}^*\}$ (i.e., we identify b with $-b$ because it doesn't matter here). Following our process from earlier, we have

$$\mathbb{R}[x] \to \mathrm{Frac}\left(\frac{\mathbb{R}[x]}{(x-a)}\right) \xhookrightarrow{\mathbb{R}-\mathrm{alg}} L,$$

where $L/\mathbb{R}$. Here, $\mathrm{Frac}(\mathbb{R}[x]/(x-a)) \cong \mathbb{R}$. For the irreducible quadratics, we have

$$\mathbb{R}[x] \to \mathrm{Frac}\left(\frac{\mathbb{R}[x]}{((x-a)^2+b^2)}\right) \xhookrightarrow{\mathbb{R}-\mathrm{alg}} L,$$

where $\mathrm{Frac}(\mathbb{R}[x]/((x-a)^2+b^2)) \cong \mathbb{C}$. Thus, $|\mathrm{Spec}\,\mathbb{R}[x]|$ is given by $\mathbb{C}$ modded out by complex conjugation, with $\{0\}$ added in.

Now, consider

$$\mathbb{R}[x] \to \mathrm{Frac}(\mathbb{R}[x]/(0)) \hookrightarrow L,$$

where X satisfies no algebraic equation over $\mathbb{R}$.

Day 3: Zariski Topology (Sep. 15, 2026)

It turns out that Hunter Spink types out his lecture notes! Regardless, I'll try to keep up here (until I burn out).

Recall that we defined $|\operatorname{Spec} R|$ to be the collection of prime ideals in R. For example, $|\operatorname{Spec} \mathbb{R}[x]|$ is given by $\mathbb{C}$ modded out by complex conjugation, along with the $\{0\}$ ideal itself (we call this the transcendental solution to $Z(\varnothing)$).

Recall that $\operatorname{Hom}_{\mathbb{R}-\text{alg}}(C^0(M,\mathbb{R}),\mathbb{R}) = \{\text{ev}_p : p \in M\}$. The idea being, if you want to identify $Z(f)$, we can simply look at all the points p such that $\text{ev}_p\, f = 0$ instead. Specifically, we want to look at the bijection between $\{\text{ev}_p : p \in M\}$ and $\{p \in M\}$.

Consider $|\operatorname{Spec} K|$, where K is a field. Clearly, $|\operatorname{Spec} K| = \{(0)\}$. Using the fact that $K(p) = \operatorname{Frac}(R/p)$ and $f \in R$, we have that $f(p)$ is the inverse of f in $K(p)$, i.e., $\text{ev}_p : R \to K(p)$.

Definition 3.1. For $\mathcal{A} \subset R$, we define $Z(\mathcal{A})$ to be

$$Z(\mathcal{A}) = \{p \in \operatorname{Spec} R \mid f(p) = 0, \forall f \in \mathcal{A}\} \subset \operatorname{Spec} R.$$

Specifically, the above definition is equivalent to $p \in \operatorname{Spec} R$ such that $\mathcal{A} \subset p$. Think of the above as "all functions that vanish at p".

We define the Zariski topology on $|\operatorname{Spec} R|$ to have $Z(\mathcal{A})$ as the closed sets. This is a *tentative definition*, and will be expanded upon later. We want to show that this is actually a topology!

Theorem 3.2. The Zariski topology (described above) is a topology.

Proof. First, we want to show that $\varnothing$ and $|\operatorname{Spec} R|$ lie in the Zariski topology. For $\varnothing$, we can take $Z(R)$, or really just $Z(1)$ (1 being picked because it is a unit). 1 has the great feature that it never vanishes in any field. For $|\operatorname{Spec} R|$, we can take $Z(0) = Z(\varnothing)$.

For finite unions (since we're prescribing closed, not open sets), we have that $Z(\mathcal{A}) \cup Z(\mathcal{B}) = Z(\mathcal{A}\mathcal{B})$. For arbitrary interscetions, we have

$$\bigcap Z(A_\beta) = z\left(\bigcup A_\beta\right).$$

Now, we want to check that

$$Z(\mathcal{A}\mathcal{B}) = \{p \mid fg \in p, \forall f \in \mathcal{A}, g \in \mathcal{B}\},$$

and

$$Z(\mathcal{A}) \cup Z(\mathcal{B}) = \{p \mid f \in p, \forall f \in \mathcal{A}\} \cup \{p \mid g \in p, \forall g \in \mathcal{B}\},$$

in order to check that our statement for finite unions holds; indeed, this follows from the fact that, if there exists $f \in \mathcal{A}$ with $f \notin p$ and $g \in B$ with $g \notin p$, then $fg \notin p$. □

We want to look at some examples of topological spaces now. As an example, consider $\operatorname{Spec} \mathbb{Z}$. This space has one point for every prime ideal in $\mathbb{Z}$, of which are the zero ideal and ideals generated by primes. What does $Z(4)$ look like here? $\overline{4} = 0$ in $(2) = \mathbb{F}_2$, $\overline{4} = 1$ in $(3) = \mathbb{F}_3$, and so on, with it simply being 4 in $(0) = \mathbb{Z}$. Thus, $Z(4) = \{(2)\}$ (i.e., check the prime factorization).[2] We can try this with other numbers too, such as $Z(12) = \{(2),(3)\}$, $Z(1) = \varnothing$, and $Z(0) = \operatorname{Spec} \mathbb{Z}$.

The closed sets here are $\varnothing$, $\operatorname{Spec} \mathbb{Z}$, and finite collections of primes. In particular, $\overline{\{(0)\}} = \operatorname{Spec} \mathbb{Z}$. At this point, Hunter made a remark that the facts we know about manifold topology don't apply here at all, which makes sense, since this space is... really weird.

[2] four, four, four, four... (draws a lot of 4s on the blackboard). JHIN!!!!!

Recall that

$$|\operatorname{Spec} \mathbb{R}[x]| = \left(\frac{\mathbb{C}}{z \leftrightarrow \overline{z}} \right) \cup \{(0)\}.$$

Here, $Z(f) = Z(f_1) \cup \cdots \cup Z(f_k)$, given that we can factor f as $f_1 f_2 \dots f_k$ with each f_i irreducible. For irreducible f, if $f = x - a$, then $Z(f) = \{(x - a)\}$; if $f = (x - a)^2 + b^2$, then $p = (f)$. Both $\mathbb{Z}$ and $\mathbb{R}[x]$ are UFDs; in particular, they are Dedekind domains, i.e., all nonzero proper ideals can be factored into a product of prime ideals.

We're now going to do an example where the ring is not a UFD. Consider $|\operatorname{Spec} \mathbb{C}[x, y]|$ (where $\mathbb{C}[x, y]$ is a UFD); per the Nullstellensatz, there are three types of primes. We can think of the dimension 0, 1, and 2 varieties, which will correspond to these prime ideals. We classify by height of primes, i.e., the transcendence degree of $\kappa(p)/\mathbb{C}$. For dimension 0 varieties, we can pick out individual points (a, b) of the affine space $\mathbb{A}^2$. For f irreducible, $\overline{\{(f)\}} = (f) \cup \{(x - a, y - b) \mid f(a, b) = 0\}$. "As a topological space, this thing is a disaster."

When does $Z(g)$ contain (f)? This occurs when $g = 0$ in $\operatorname{Frac}(\mathbb{C}[x, y]/(f))$, which is equivalent to $g \in (f)$. Then g is a multiple of f. Since $\mathbb{C}[x, y]$ is a UFD, the smallest $Z(s)$ contains f if $Z(f)$ is given by p such that $f(p) = 0$, i.e., $\{(x - a, y - b), f(x, y) = 0, (f)\}$. You can always find closures from first principles, but to be more efficient you need "non-tautologous math".

Day 4: Spectrum Properties (Sep. 17, 2026)

Recall, from last time, that we took

$$|\operatorname{Spec} R| = \{p \in R \mid p \text{ is prime}\},$$

and gave it a topology with a basis of closed sets $Z(f) = \{p \in \operatorname{Spec} R \mid f(p) = 0\}$, where $f \in R$, and $f(p) = 0$ meant the image of f in $k(p) = \operatorname{Frac}(R/p)$. Equivalently, $f \in p$.

Theorem 4.1. If $I = (A)$ is the ideal generated by $A \subset R$, then $z(A) = z(I) = \{p \in \operatorname{Spec} R \mid I \subset p\}$.

Proof. $z(A) = \{p \in \operatorname{Spec} R \mid A \subset p\} = \{p \in \operatorname{Spec} R \mid I \subset p\} = z(I)$. □

Theorem 4.2. The assignment $R \to |\operatorname{Spec} R|$ is a contravariant functor **Ring** $\to$ **Top**.

We introduce some category theory here. A *category* is a set of objects $\operatorname{Ob}(C)$, a collection of morphisms $\{\operatorname{Mor}(X, Y) : X, Y \in \operatorname{Ob}(C)\}$, and a composition map

$$\circ\colon \operatorname{Mor}(X, Y) \times \operatorname{Mor}(Y, Z) \to \operatorname{Mor}(X, Z)$$

satisfying the properties,

(i) Composition is associative, i.e., $f \circ (g \circ h) = (f \circ g) \circ h$ whenever this makes sense,

(ii) An identity map exists, i.e., $\mathrm{id}_X \in \operatorname{Mor}(X, X)$ such that $\mathrm{id}_X \circ f = f = f \circ \mathrm{id}_X$ whenever $f \in \operatorname{Mor}(X, Y)$.

In the theorem above, there are two categories: rings and topological spaces. **Ring** is a category whose objects are rings and morphisms are ring maps (there's nothing to check here because the morphisms are determined at the level of elements). In particular, **Ring** is a subcategory of **Set**. **Top** is a category whose objects are topological spaces and morphisms are continuous maps.

A *functor* $F\colon \mathscr{C} \to \mathscr{D}$ between categories is the data of two maps (both denoted F)

$$F\colon \operatorname{Ob}(\mathscr{C}) \to \operatorname{Ob}(\mathscr{D}), \quad F\colon \operatorname{Mor}(X, Y) \to \operatorname{Mor}(F(X), F(Y)),$$

satisfying the following relations,

(i) $F(f) \circ F(g) = F(f \circ g)$ when this makes sense,

(ii) $F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$.

We now define the notion of an *opposite category*. Let $\mathscr{D}$ be a category. Then $\mathscr{D}^{\mathrm{op}}$ as a category consists has $\operatorname{Ob}(\mathscr{D}^{\mathrm{op}}) = \operatorname{Ob}(\mathscr{D})$ and

$$\operatorname{Mor}_{\mathscr{D}^{\mathrm{op}}}(X, Y) \coloneqq \{f^{\mathrm{op}} \mid f \in \operatorname{Mor}_{\mathscr{D}}(Y, X)\}.$$

Composition works by $f^{\mathrm{op}} \circ g^{\mathrm{op}} \coloneqq (g \circ f)^{\mathrm{op}}$. With this, we may now define contravariant functors. A *contravariant functor* is a functor $F\colon \mathscr{C} \to \mathscr{D}^{\mathrm{op}}$ with

$$F\colon \operatorname{Ob}(\mathscr{C}) \to \operatorname{Ob}(\mathscr{D}), \quad F\colon \operatorname{Mor}(X, Y) \to \operatorname{Mor}(F(Y), F(X)),$$

satisfying obvious rules $F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$ and $F(f \circ g) = F(g) \circ F(f)$. As an example of more functors, we have the forgetful functor **Ring** $\to$ **Set**, which is basically a functor that "forgets the ring structure". We also have the contravariant functor from $\mathbf{Vect}_K \to \mathbf{Vect}_K$, where we sent $V \to V^* = \operatorname{Hom}(V, K)$, given by the dual map $(f\colon V \to W) \mapsto (f^*\colon W^* \to V^*)$.

As another example, consider the functor **Mfld** $\to$ **Ring**, given by $F\colon M \to C^0(M,\mathbb{R})$, and $(\Phi\colon M \to N) \mapsto (\Phi^*\colon C^0(N,\mathbb{R}) \to C^0(M,\mathbb{R}))$, i.e., the pullback.

We now prove 4.2.

Proof. Given a map $f\colon R \to S$, we can define $f^*\colon \operatorname{Spec} S \to \operatorname{Spec} R$ by $q \mapsto f^{-1}(q)$. We want to claim that $f^{-1}(q)$ is actually a prime ideal. Suppose $ab \in f^{-1}(q)$; equivalently, $f(ab) \in q$; since f is a ring map, this is equivalent to saying that $f(a) \in q$ or $f(b) \in q$, so $a \in f^{-1}(q)$ or $b \in f^{-1}(q)$. Thus, $f^{-1}(q) \in \operatorname{Spec}(R)$.

We now want to check that f^* is a continuous map of topological spaces. Take a closed set $z(I) \subset \operatorname{Spec}(R)$; we want to look at $(f^*)^{-1}(z(I))$. By definition,

$$(f^*)^{-1}(z(I)) = \{q \in \operatorname{Spec} S \mid f^*(q) \in z(I)\}.$$

$f^*(q)$ is $f^{-1}(q)$, so $f^*(q) \in z(I)$ is equivalent to $I \subset f^{-1}(q)$, which is the same as $f(I) \subset q$. Thus, the above set is simply $z(f(I))$.

Finally, we want to check that we actually have a contravariant functor. This is really easy, since the map we have is defined by set maps. First, $(\mathrm{id}_R)^* = \mathrm{id}_{\operatorname{Spec} R}$; second, if we have a map $g\colon R \to S$ and $f\colon S \to T$, then $f^*\colon \operatorname{Spec} T \to \operatorname{Spec} S$ and $g^*\colon \operatorname{Spec} S \to \operatorname{Spec} R$, and we have $(f \circ g)^* = g^* \circ f^*$ as desired. Indeed, for a prime ideal $t \subset T$, we have $(f \circ g)^*(t) = (f \circ g)^{-1}(t) = g^{-1} \circ f^{-1}(t) = g^* \circ f^*(t)$. □

"Yay, we've created half the definition for an affine scheme so far."

Theorem 4.3. Suppose $\Phi\colon R \to S$. Then for $f \in R$, we have $\Phi(f) = f_{|\operatorname{Spec} R|} \circ \Phi^*$, where we consider $f\colon \operatorname{Spec} R \to \prod_{p \in \operatorname{Spec}(R)} k(p)$.

$$\begin{array}{ccc} S & \xleftarrow{\Phi(f)} & R \\ \vdots & & \vdots \\ \operatorname{Spec} S & \xrightarrow{\Phi^*} & \operatorname{Spec} R \end{array}$$

Proof. Hunter said he would leave this as an exercise once he figures out the statement (why...). □

Theorem 4.4. Let $Y \subset \operatorname{Spec} R$. Then $\overline{Y} = Z\left(\bigcap_{p \in Y} p\right)$,[3]

Proof. $\overline{Y} = z(I)$. Suppose $z(J) \supset Y$; equivalently, J lies in each prime ideal of Y, which we can write as

$$J \subset \bigcap_{p \in Y} p.$$ □

Corollary 4.5. $\overline{\{p\}} = z(p) = \{q \in \operatorname{Spec} R \mid p \subset q\}$.

Intuitively, "p is a generic point inside its closure $\overline{\{p\}}$". As an example from last class, if we look at $\operatorname{Spec} \mathbb{C}[x,y]$, then $\overline{\{f\}}$ is given by the points on the curve $\bigcup\{(f)\}$. He then said something about the map $\mathbb{C}[x,y]/(f) \to \operatorname{Frac}(\mathbb{C}[x,y]/(f))$, but I didn't catch it.

Corollary 4.6. If $\overline{z} \subset z(p)$, then every open set containing $\overline{z}$ contains p.

As a corollary to our discussion, a prime ideal p is closed if and only if p is maximal.

[3] i swear to god if i should've been using $\mathfrak{p}$ instead of p this whole time i will cry

Let's look at an example of an interesting ring that'll we'll see a lot in the future. Recall that if K is a field, then Spec K is a single point $\{(0)\}$. The next most complicated space, as spectrums go, is a local ring.

Definition 4.7. A *local ring* is a ring with exactly one maximal ideal.

Corollary 4.8. Spec R has a unique closed point if R is a local ring.

As an example, $K[X]_{(x-5)} = \{\frac{f}{g} : f, g \in K[X], (x-5) \text{ divides } f \text{ as many times as it divides } g\}$. $(x-5) \subset K[X]_{(x-5)}$ is a maximal ideal because the kernel of the evaluation map $\text{ev}_5 \colon K[X]_{(x-5)} \to K$ is $(x-5)$. Note that, if $f(5) \neq 0$, then f is a unit. Now,

$$\frac{f}{g} = (x-5)^a \frac{\tilde{f}}{\tilde{g}},$$

where $\tilde{f}$ and $\tilde{g}$ don't vanish at 5. Indeed, $\frac{1}{f} \in K[X]_{(x-5)}$.

Given any ideal $I \subset K[X]_{(x-5)}$ generated by $\mathcal{A}$, replace each element of $\mathcal{A}$ with a power of $x-5$; then I is generated by some list of powers of $x-5$. This means $I = (x-5)^a$, or $I = (a)$, meaning if $a \geq 2$, then it is not prime, since $(x-5)(x-5)^{a-1} \in ((x-5)^a)$. In this manner, in

$$\text{Spec } K[X]_{(x-5)} = \{(0), (x-5)\},$$

the closed sets are (0) (i.e., the whole space), $(x-5)$, and $\varnothing$.

We then gave a small preview of the structure sheaf. We now end with a final "easy observation".

Theorem 4.9. Spec R/I embeds into Spec R with the subspace topology as $z(I)$.

"Next week, we're going to do the real meat and potatoes of... spectrums.'

Day 5: Localization (Sep. 22, 2026)

Given a function $\phi\colon R \to S$ (thought of as function pullback), we have that $\phi^*\colon |\operatorname{Spec} S| \to |\operatorname{Spec} R|$ and $(\phi^*)^{-1}Z(I) = Z(\phi(I))$.

Definition 5.1. Given $f \in R$, define the basic open subset

$$D(f) \coloneqq Z(f)^c = \{\mathfrak{p} \in \operatorname{Spec} R \mid f(\mathfrak{p}) \neq 0\}.$$

The $D(f)$'s form a subset of the topology; i.e., every open set is a union of these $D(f)$, and we have

$$D(f_1) \cap D(f_2) \cap \cdots \cap D(f_k) = D(f_1 \dots f_k).$$

Theorem 5.2 (Rabinowitsch)**.** Let $f_1, \dots, f_n, g \in k[x_1, \dots, x_n]$; then

$$Z_k(f_1, \dots, f_n) \cap D_k(g) = Z_k(f_1, \dots, f_n, x_{n+1}g - 1).$$

Specifically, the left hand side is a subset of k^n, and the right hand side is a subset of k^{n+1}.

Proof. The proof is simple; consider the correspondence given by the two maps

$$(a_1, \dots, a_n) \mapsto \left(a_1, \dots, a_n, \frac{1}{g(a_1, \dots, a_n)}\right), \quad (b_1, \dots, b_n) \leftarrow\!\shortmid (b_1, \dots, b_{n+1}). \qquad \square$$

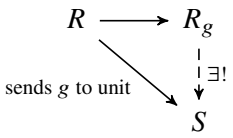
Definition 5.3. We define the localization of R at g as $R_g \coloneqq R[x]/(xg - 1)$. We will denote x as g^{-1} freely.

Theorem 5.4. We have the following properties for localization,

(i) Every element of R_g can be written as rg^{-1} with $r \in R$.

(ii) $rg^{-1} \in Sg^{-1}$ if and only if $g^N(r - s) = 0$ for some N.

(iii) A ring map $R_g \to S$ is the same as a map $R \to S$ such that g maps to a unit. Specifically, this is a universal property



(iv) This map from $R \to R_g$ is an isomorphism if and only if g is a unit. In words, "if you tried to localize at a unit, you didn't actually do anything".

Proof. For (i), take some element of the localization R_g, and write it as $f(x) = r_0 + r_1x + \cdots + r_nx^n$. We can factor out x^n by writing

$$f(x) = (r_0g^n + \cdots + r_1g^{n-1} + \cdots + r_n)x^n,$$

for which the bracketed portion is in R, and $x^n = g^{-n}$, so we are done.

For (ii), by taking the difference of two sides, it suffices to show that $r = 0$ in R_g if and only if $g^Nr = 0$ in R, or $r = 0$ in R_g. We really just want to look at the kernel of the map from $R \to R_g$, consisting of the elements which are annihilated by some power of g. If $r \mapsto 0$ in $R[x]/(xg - 1)$, then in $R[x]$, $r = (1 - xy)(a_0 + a_1x + \cdots + a_nx^n)$; by comparing coefficients, we get

$$r = a_0 + (a_1 - ga_0)x + (a_2 - ga_1)x^2 + \cdots + (a_n - ga_n{-}1)x^n - a_ngx^{n+1}.$$

This tells us that $a_0 = r$, $a_1 = gr$, $a_2 = g^2 r$, etc., so $a_n = g^n r$, which we know is 0. The converse is obvious: if $g^N r = 0$ in R, then $g^N r = 0$ in R_g, i.e., $r = 0$ in R_g.

For (iii), a map from $R[x]/(xg - 1) \to S$ is the same as $x \mapsto s$ where $sg = 1$.

For (iv), if g is a unit in R, then $R[x]/(xg - 1)$ can be written as $R[x]/(x - g^{-1}) \cong R$. □

As an example, $\mathbb{Z}_0 = 0$. This is because $\mathbb{Z}[x]/(x(0) - 1) = \mathbb{Z}[x]/(1) = 0$. More generally, if g is nilpotent, i.e., $g^N = 0$, then $R_g = 0$, since $g^N r = 0$. I kinda got lost at these examples cuz the board was too dusty.

Theorem 5.5. $\left|\operatorname{Spec} R_g\right| \cong D(g) \subset |\operatorname{Spec} R|$.

Proof. Consider the map $\phi\colon R \to R_g$, so $\phi^*\colon \left|\operatorname{Spec} R_g\right| \to |\operatorname{Spec} R|$. First, we can show that $\operatorname{Im} \phi^* \subset D(g)$. Indeed, $(\phi^*)^{-1}(D(g)) = Z(\phi(g))$ (since zero sets pullback the way you expect), which is equal to ϕ.

Now, we want to show that ϕ^* is injective, i.e., we can recover $\mathfrak{p}$ from $\phi^*\mathfrak{p} = \phi^{-1}(\mathfrak{p})$. We claim that $(\phi(\phi^{-1}(\mathfrak{p}))) = \mathfrak{p}$. The non-trivial containment is that $\mathfrak{p} \subset \phi(\phi^{-1}(\mathfrak{p}))$. Given r/g in $\mathfrak{p}$, we have $r \in \mathfrak{p}$, so $r \in \phi^{-1}(\mathfrak{p})$, so $r \in \phi(\phi^{-1}(\mathfrak{p}))$, whence $r/g \in (\phi(\phi^{-1}(\mathfrak{p})))$.

It remains to check that ϕ^* is surjective. Take $\mathfrak{q} \in D(g) \subset \operatorname{Spec} R$, and note that $R \to \kappa(\mathfrak{q})$ sends g to a unit. Thus, we can factor the map as $R \to R_g \to \kappa(\mathfrak{q})$. Taking the spectrum of this factorization, we get $\operatorname{Spec} R \leftarrow \operatorname{Spec} R_g \leftarrow \operatorname{Spec} \kappa(\mathfrak{q})$, so $\operatorname{Spec} R_g$ contains $\mathfrak{q}$.

Finally, given $D(h) \in \operatorname{Spec} R_g$, we want to show that $D(h)$ is open in $\operatorname{Spec} R_g$. We have that $h = f/g^i$ for some i and $f \in R$; for $\phi\colon R \to R_g$, without loss of generality, $i = 0$, so $h = \phi(f)$ in R_g. Thus,

$$(\phi^*)^{-1} D(f) = D(\phi(f)) = D(h);$$

but $(\phi^*)^{-1}$ is just restriction, so $D(h) = D(f)$ is open in $\operatorname{Spec} R$. □

He will correct this proof next class, apparently. "iunno. fix this proof. it's an exercise"

Day 6: Localization and Germs (Sep. 24, 2026)

Last class, we saw that, for $\phi\colon R \to R_g$, we have that $\phi^*\colon \operatorname{Spec} R_g \hookrightarrow \operatorname{Spec} R$ has image $D(g) \subset \operatorname{Spec} R$. We want to show that it is homeomorphism onto its image. Given an open set $D(f/g^i) \subset \operatorname{Spec} R_g$, we want an open set U in $\operatorname{Spec} R$ such that $(\phi^*)^{-1}U = D(f/g^i)$. Indeed, $D(f/g^i)_{\operatorname{Spec} R_g} = D(f)$, so we may take $U = D(f) \subset \operatorname{Spec} R$.

"By now, we're getting *really* close to the definition of a scheme... next week, we'll be 'full scheme ahead'."

Definition 6.1. An ideal $I \subset R$ is said to be *radical* if $I = \sqrt{I} \coloneqq \{r \in R \mid r^n \in I \text{ for some } n\}$. $\sqrt{I}$ is called the *radical* of I.

Note that the radical of zero, $\sqrt{(0)}$, is equal to the set of all nilpotent elements of R, which we see by unraveling the definitions. Recall that $f(\mathfrak{p}) = 0$ for all $\mathfrak{p} \in \operatorname{Spec} R$ if f is nilpotent. Moreover, I being radical is equivalent to R/I being reduced, i.e., it does not have any nilpotent elements. In general, $(R/I)_{\text{red}} \coloneqq (R/I)/\sqrt{(0)} = R/\sqrt{I}$.

Theorem 6.2. Let $I \subset R$ be an ideal; then

$$\sqrt{I} = \bigcap_{\mathfrak{p} \supset I} \mathfrak{p}.$$

In particular, $\sqrt{(0)}$ is the intersection of all prime ideals $\mathfrak{p} \in \operatorname{Spec} R$, i.e., $f(\mathfrak{p}) = 0$ for all $\mathfrak{p}$ if and only if f is nilpotent.

Proof. By quotienting by I, we can reduce the question to the case $I = (0)$. We know

$$\sqrt{(0)} \subset \bigcap_{\mathfrak{p} \in \operatorname{Spec} R} \mathfrak{p}$$

because if r is nilpoetnt, then $r^n = 0 \in \mathfrak{p}$ implies $r \in \mathfrak{p}$. Conversely, we want to show that if $f \in \bigcap_{\mathfrak{p} \in \operatorname{Spec} R} \mathfrak{p}$, then f is nilpotent. Indeed, $D(f) = \varnothing$, i.e., there is no ideal in which f does not vanish in. Since $D(f) = \operatorname{Spec} R_f$, we see that this means R_f is the zero ring, i.e., in R_f, we have $1 = 0$. Thus, in R, $f^n \cdot 1 = 0 f^n$ for some n, and so $f^n = 0$. □

Note that we implicitly use Zorn's lemma above, i.e., if $\operatorname{Spec} R_f = \varnothing$, then R_f is the zero ring.

Theorem 6.3. $Z(I) = Z(\sqrt{I})$, and Z gives a contravariant equivalence.

$$\{\text{radical ideals in } R\} \to \{\text{closed subsets of } \operatorname{Spec} R\}.$$

Proof. For the first part, $Z(I) = Z(\bigcap_{\mathfrak{p} \in I} \mathfrak{p}) = Z(\sqrt{I})$. For the second part, we ask, how do we recover the radical I from the zero locus? Since $Z(I) = \{\mathfrak{p} \mid \mathfrak{p} \supset I\}$, we see that

$$\bigcap_{\mathfrak{p} \in Z(I)} \mathfrak{p} = \sqrt{I} = I. \qquad \square$$

Given a closed set V, we may define $I(V) = \bigcap_{\mathfrak{p} \in V} \mathfrak{p}$, and we have $Z(I(V)) = V$ and $I(Z(J)) = \sqrt{J}$.

Corollary 6.4. We have the following,

(i) $D(f) = \varnothing$ if and only if $Z(f) = \operatorname{Spec} R$ if and only if f is nilpotent.

(ii) $D(f) = \operatorname{Spec} R$ if and only if $Z(f) = \varnothing$ if and only if f is a unit.

(iii) $\operatorname{Spec} R = \bigcup_{\alpha\in\mathcal{A}} D(f_\alpha)$ if and only if $Z(\{f_\alpha \mid \alpha \in \mathcal{A}\}) = \varnothing$ if and only if $\{f_\alpha \mid \alpha \in \mathcal{A}\} = (1)$,

As a throwback to day 1 of class, observe that $x^2 + 1$ does not vanish on any real number, but it is also not invertible. We may look at the prime ideal $\mathfrak{p} = (x^2 + 1)$ in $\mathbb{R}[x]$ to see $x^2 + 1 = 0$ in $\kappa(\mathfrak{p})$, so (ii) does not apply. Also note that, in classical algebraic geometry, the Nullstellensatz states that if $k = \overline{k}$, and $f_1, \ldots, f_r$ have no common solution, then there exists $g_!, \ldots, g_r$ such that $\sum_i g_i f_i = 1$.

Proof. We just proved (i) earlier. For (ii), we have that $I(Z(f)) = \sqrt{(f)}$, so $I(\varnothing) = (1)$, then $1^n \in (f)$ for some n, which tells us $1 \in (f)$, whence f is a unit. For (iii),

$$I(Z(\{f_\alpha \mid \alpha \in \mathcal{A})) = \sqrt{(f_\alpha \mid \alpha \in \mathcal{A})},$$

and the left hand side is equal to $I(\varnothing) = (1)$, so $1^n \in (f_\alpha \mid \alpha \in \mathcal{A})$, whence $(f_\alpha \mid \alpha \in \mathcal{A}) = (1)$. □

Corollary 6.5. $\operatorname{Spec} R$ is quasi-compact.

Proof. By (iii) from 6.4, we have that $(f_\alpha \mid \alpha \in \mathcal{A}) = (1)$, so there exists $\sum_{i=1}^n f_i g_i = 1$, which implies that

$$\operatorname{Spec} R = \bigcup_{i=1}^{n} D(f_i).$$ □

In this manner, we may think $\operatorname{Spec} R_p$ "=" $D(f) \subset \operatorname{Spec} R$. The problem with writing equality is that, if we had another function g such that $D(f) = D(g)$, then we might want to have a different canonical ring to describe $D(g)$. For example, $D(x) = D(x^2)$, and $\operatorname{Spec} \mathbb{C}[x]_x = Z(xy - 1)$, and $\operatorname{Spec} \mathbb{C}[x]_{x^2} = Z(x^2y - 1)$.

Theorem 6.6. If $D(f) = D(g)$, then there exists a unique R-algebra map between R_f and R_g. In particular, this map is an isomorphism.

Proof. Look at $Z(f)_{\operatorname{Spec} R_g} = \varnothing$. Then f is a unit in R_g, so there is a canonical map $R_f \to R_g$, and vice versa. □

Corollary 6.7. For a basic open set U, the "ring of functions" is R localized at *all* $f \in R$ such that f does not vanish on U.

Theorem 6.8. The localization of a localization is the localization of the product, i.e., $(R_f)_g = R_{fg}$. Another way to think about this is to say that restricting from $D(f)$ to $D(g)_{D(f)}$ is the same as directly restricting to $D(fg)$.

Proof. The proof is left as a homework exercise. □

We now talk about germs of functions. A germ is a function defined on a neighborhood of a point; specifically, we define a *germ* at $\mathfrak{p} \in \operatorname{Spec} R$ to be a pair (f, U) (where $U \ni \mathfrak{p}$ is a basic open set) such that $f \in R_g$ up to an equivalence relation that $(f_1, U_1) \sim (f_2, U_2)$ provided there exists a basic open set $U_3 \subset U_1 \cap U_2$ such that $\mathfrak{p} \in U_3$ and $(f_1|_{U_3}, U_3) = (f_2|_{U_3}, U_3)$. We may regard $f_1|_{U_3}$ as the inverse of f in $\mathcal{O}(U_3)$.

He then gave an example with parabolas on the board.

Definition 6.9. The localization of R at $\mathfrak{p}$ is given by $R_{\mathfrak{p}} \coloneqq R[\{x_f \mid f \notin \mathfrak{p}\}]/\{1 - x_f f \mid f \notin \mathfrak{p}\}$.

Theorem 6.10. R_p is isomorphic to the ring of germs of R at p.

Proof. We can map R_p to the ring of germs by $r \mapsto (r, \operatorname{Spec} R)$ and $x_f \mapsto (f^{-1}, D(f))$. Clearly, this is well-defined based on how we wrote the definitions earlier. Surjectivity is also obvious. What about injectivity, though? Suppose we have some element of R_p. This only involves finitely many of the formal inverses, so we can collect denominators, so we may write r/g. Suppose $r/g \mapsto 0$, i.e., $(r/g, D(g)) = 0$. This means, there exists h such that $D(h) \subset D(g)$, so that $d/g|_{D_h)} = 0$. Equivalently, $r|_{D(h)} = 0$, so $h^N r = 0$ in R, so $r = 0$ in R_p, as this is a unit in R, so $r/g = 0$ in R. □

Day 7: Ideals of Localization and Sheaf properties (Sep. 29, 2026)

Theorem 7.1. We have the following properties of localization,

(i) $\mathfrak{p}R_\mathfrak{p}$ is the *unique* maximal ideal (i.e., $R_\mathfrak{p}$ is a local ring).

(ii) $\left|\operatorname{Spec} R_\mathfrak{p}\right| \cong \{\mathfrak{q}R_\mathfrak{p} \mid \mathfrak{q} \in \operatorname{Spec} R,\ \mathfrak{q} \subset \mathfrak{p}\} \subset \operatorname{Spec} R$, i.e., a homeomorphism.

Proof. $R_\mathfrak{p}$ considered as germs $(f/g^i, D(g))$ has an obvious ideal: the set of germs vanishing at $\mathfrak{p}$. This is an ideal, since, if we have a germ $(f/g^i, D(g))$ not in this ideal, then restricting further to $D(fg) = \mathfrak{p}$ yields us a unit

$$\left(\left.\frac{f}{g^i}\right|_{D(fg)}, D(fg)\right),$$

so the germ is a unit. Consequently, no proper ideal can contain an element outside of this ideal, so this ideal is maximal.

Finally, note that any element of this ideal $(f/g^i, D(g))$ has $f(\mathfrak{p}) = 0$, and hence is equivalent, up to a unit, to $(f, \operatorname{Spec} R)$ with $f \in \mathfrak{p}$. Thus, this ideal lies in $\mathfrak{p}R_\mathfrak{p}$. The converse is obvious, since, if $f \in \mathfrak{p}$, then $\mathfrak{p}R_\mathfrak{p}$ is in the ideal, and we have $f(\mathfrak{p}) = 0$.

For (ii), we claim that each prime ideal is of the form $\mathfrak{q}R_\mathfrak{p}$ for some $\mathfrak{q} \in \operatorname{Spec} R$. As shown in (i), any ideal in $R_\mathfrak{p}$ is of the form $IR_\mathfrak{p}$ for some $I \subset R$. In particular, this ideal has the further property that if $rs \in I$ and $s \notin \mathfrak{p}$, then $r \in I$. Given an ideal with this property, we want to show that $IR_\mathfrak{p}$ is prime is equivalent to $I \subset R$ being prime.

First, if $IR_\mathfrak{p}$ is prime, suppose $ab \in I$; we want to show that one of a, b is in I. Indeed, note that, if $a_\mathfrak{p} b_\mathfrak{p} \in IR_\mathfrak{p}$ (where $a_\mathfrak{p}, b_\mathfrak{p} \in R_\mathfrak{p}$), then one of $a_\mathfrak{p}$ or $b_\mathfrak{p}$ lie in $IR_\mathfrak{p}$; suppose it is $a_\mathfrak{p}$. But then $a_\mathfrak{p} \in IR_\mathfrak{p}$ implies that there exists $f \in \mathfrak{p}$ such that $fa \in I$, so $a \in I$. For the converse, we have that

$$\frac{R_\mathfrak{p}}{IR_\mathfrak{p}} \cong \left(\frac{R}{I}\right)_\mathfrak{p} \to \operatorname{Frac}(R/I).$$

Finally, given $\alpha = (f/g^i, D(g)) \in R_\mathfrak{p}$, we want to show that $D(\alpha)$ is the intersection of an open subset of $\operatorname{Spec} R$ with $\operatorname{Spec} R_\mathfrak{p}$. Indeed, $D(f) \cap \operatorname{Spec} R_\mathfrak{p} = D(\alpha)$. □

We now talk about the sheaf property. Consider $C^0(M, \mathbb{R})$, and consider the open cover $\{U_\alpha\}_{\alpha \in \mathcal{A}}$. We may evaluate continuous functions on each U_α, and we may look at continuous functions on each U_α that are compatible on each $U_\alpha \cap U_\beta$, i.e.,

$$f_\alpha\big|_{U_\alpha \cap U_\beta} = f_\beta\big|_{U_\alpha \cap U_\beta},$$

to patch together globally continuous functions on the entire of M; this is because continuity is a local property.

As an example, let us consider $\operatorname{Spec}\mathbb{Z}$. $D(9)$ and $D(5)$ cover $\operatorname{Spec}\mathbb{Z}$, since the ideal generated by 9 and 5 contain 1. In terms of the spectrum,

$$D(9) = \operatorname{Spec}\mathbb{Z} \setminus (3), \quad D(5) = \operatorname{Spec}\mathbb{Z} \setminus (5).$$

Clearly, the union is the whole of $\operatorname{Spec}\mathbb{Z}$, since each prime is either divisible 3 or 5, but not both (usually, it's divisible by neither). The intersection of $D(9)$ and $D(5)$ is $D(45) = \operatorname{Spec}\mathbb{Z} \setminus (3, 5)$.

Functions in $D(9)$ are of the form $\frac{x}{9^i}$ and in $D(5)$, they're of the form $\frac{y}{5^j}$. Assuming

$$\frac{x}{9^i} = \frac{y}{5^j} \in \mathbb{Z}_{45} \subset \mathbb{Q},$$

we have that

$$\frac{x}{9^i} = \frac{y}{5^j} = z$$

for some unique integer z. $D(9^i) = D(9)$ and $D(5^j) = D(5)$, so $(1) = (9^i, 5^j)$, so $(1) = (9, 5)^{i+j}$, and we have $1 = c9^i + d5^j$.[4] Observe that

$$\frac{x}{9^i} = \frac{y}{5^j} = \frac{cx + dy}{c9^i + d5^j} = cx + dy.$$

Let us look at a more general example. In $\mathbb{R}[x, y]$, we have that

$$Z(y - x^2) \leftrightarrow Z(y^2 - x^4 + 1) \subset Z(y - x^2) \cup Z(y^7 - x^4 + 1).$$

$Z(y - x^2)$ and $Z(y^2 - x^4 + 1)$ are disjoint; how do we glue them together? By taking $1 = (y^2 - x^4 + 1) - (y + x^2)(y - x^2)$... and then we ran out of time. Hunter will continue this example next lecture.

[4] "we now do a really weird property of fractions that's only taught in russia"

Day 8: Gluing Functions, Structure Sheaves, and Affine Schemes (Oct. 1, 2026)

Today, we will talk more about the sheaf properties of rings. In $\mathbb{Z}$, with a, b suitably chosen such that $a9^i + b5^j = 1$, we have that

$$\frac{x}{9^i} = \frac{y}{5^j} = \frac{ax + by}{a9^i + b5^j} = ax + by \in \mathbb{Z}.$$

Here, $ax + by$ is given by $\frac{x}{9^i}$ in $\mathbb{Z}_9$ and $\frac{y}{5^j}$ in $\mathbb{Z}_5$. Now consider $Z((y - x^2)(y^2 - x^4 + 1)) \subset \operatorname{Spec} \mathbb{C}[x, y]$. Can we interpret a polynomial, of whose restrictions to two different varieties, take on different values? Specifically, we want to find $h(x, y)$ restricting to $f(x, y)$ and $g(x, y)$ on $Z(y - x^2)$ and $Z(y^2 - x^4 + 1)$ respectively. Consider

$$1 = (y^2 - x^4 + 1) - (y + x^2)(y - x^2). \tag{†}$$

If we look at $Z(y^2 - x^4 + 1) = D(y - x^2)$, we have that

$$\left(\frac{\mathbb{C}[x, y]}{(y - x^2)(y^2 - x^4 + 1)}\right)_{y-x^2} \cong \left(\frac{\mathbb{C}[x, y]}{(y^2 - x^4 + 1)}\right)_{y-x^2},$$

since, under the localization, $y - x^2$ becomes a unit. Similarly, $Z(y - x^2) = D(y^2 - x^4 + 1)$. Vacuously,

$$\frac{f}{1} = \frac{g}{1}$$

on $D(y - x^2) \cap D(y^2 - x^4 + 1) = D((y - x^2)(y^2 - x^4 + 1)) = \varnothing$. In order to do the cross-multiplication trick (as seen in $\mathbb{Z}$), we can look at

$$\frac{f(y^2 - x^4 + 1)}{y^2 - x^4 + 1} \text{ “}=\text{” } \frac{g(y - x^2)}{y - x^2} \quad \text{in} \quad \frac{\mathbb{C}[x, y]}{(y - x^2)(y^2 - x^4 + 1)},$$

where the quoted equal sign means the grade school equality $\frac{a}{b} = \frac{c}{d}$ holding if $ad = bc$. Looking at the linear combination we created in (†), observe that the above is equal to

$$\cdots = \frac{f(y^2 - x^4 + 1) - (y + x^2)g(y - x^2)}{y^2 - x^4 + 1 - (y + x^2)(y - x^2)} = (y^2 - x^4 + 1)f - (y + x^2)(y - x^2)g.$$

We now ask, does the latter expression restrict correctly to $\mathbb{C}[x, y]/(y - x^2)$ and $\mathbb{C}[x, y]/(y^2 - x^4 + 1)$? In $\mathbb{C}[x, y]/(y - x^2)$, it is clear that the g term vanishes; since $(y^2 - x^4 + 1)$ factors as seen in (†), it follows that we get g. In $\mathbb{C}[x, y]/(y^2 - x^4 + 1)$, we may once again follow the same process (and check the equation in (†)) to get a restriction down to g as desired.

“While this isn’t super rigorous math, it gives an idea of what we’re trying to do anyways.” – Hunter

Theorem 8.1. Given an open cover $D(f_1) \cup \cdots \cup D(f_k)$ of $\operatorname{Spec} R$ and $g_i \in R_{f_i}$ such that

$$g_i|_{D(f_i f_j)} = g_j|_{D(f_i f_j)}$$

for all i, j, there exists a *unique* $g \in R$ such that $g|_{D(f_i)} = g_i$.

In the above, we have that $R \to \prod R_{f_i} \to \prod R_{f_i f_j}$ is dual to the notion $U \times_{\operatorname{Spec} R} U \to U \to \operatorname{Spec} R$, for $U = U_1 \sqcup U_2 \sqcup \cdots \sqcup U_k$.

Proof. Write $g_i = \frac{r_i}{f_i^N}$ for some common N. We know that $g_i = g_j$ on $D(f_i f_j)$ for all i, j, so there exists some M_{ij} such that

$$(f_i f_j)^M \left(\frac{r_i}{f_i^N} - \frac{r_j}{f_j^N} \right) = 0.$$

Specifically, $f_i^{M-N} f_j^M r_i = f_i^M f_j^{M-N} r_j$ in R. Replace r_i with $f_i^{M-N} r_i$ and f_i^N with f_i^M everywhere. In this manner, we can assume that $r_i f_j^N = r_j f_i^N$ for all i, j, so we may actually cross multiply.

$D(f_i) = D(f_i^N)$, so $D(f_i^N) \cup \cdots \cup D(f_k)^N) = \operatorname{Spec} R$, meaning

$$(1) = (f_1^N, \ldots, f_k^N) \implies 1 = a_1 f_1^N + \cdots + a_k f_k^N.$$

We are now going to "ansatz a way that they can glue together". Indeed, choose $r = a_1 r_1 + \cdots + a_k r_k$. When restricting r to the open set $D(f_i)$, we need to check that

$$g\big|_{D(f_i)} = g_i = \frac{r_i}{f_i^N} \iff \frac{a_1 r_1 + \cdots + a_k r_k}{1} = \frac{r_i}{f_i^N} \text{ in } D(f_i).$$

From here, we may immediately check that

$$a_1 f_i^N r_1 + a_2 f_i^N r_2 + \cdots + a_k f_i^N r_k = a_1 f_1^N r_i + a_2 f_2^N r_i + \cdots + a_k f_k^N r_i = r_i.$$

We now check uniqueness. We want to show that, if $r \in R$ and $r|_{D(f_i)} = 0$, then $r = 0$. Indeed, there exists M such that $f_i^M r = 0$ in R for all i. Then

$$D(f_1) \cup \cdots \cup D(f_k) = D(f_1^M) \cup \cdots \cup D(f_k^M) = \operatorname{Spec} R,$$

for which $1 = a_1 f_1^M + \cdots + a_k f_k^M$ imply that $r = a_1 f_1^M r + \cdots + a_k f_k^M r = 0$. □

Corollary 8.2. Given a basic open covering $\bigcup_{i=1}^n D(f_i)$ of a basic open set $D(f)$ and $g_i \in R_{f_i}$ such that $g_i = g_j$ on $D(f_i f_j)$ for all i, j, then there exists a unique $g \in R_f$ such that $g|_{D(f_i)} = g_i$.

Proof. Apply 8.1 to R_f. □

Definition 8.3. We denote $\mathcal{O}_{\operatorname{Spec} R}(U)$ to be the ring of regular functions on an open set $U \subset \operatorname{Spec} R$. It is defined as the set of tuples

$$f(\mathfrak{p})_{\mathfrak{p} \in U} \in \prod R_{\mathfrak{p}}$$

such that, for all $\mathfrak{p} \in U$, there exists a basic open set $\mathfrak{p} \in D(g_{\mathfrak{p}}) \subset U$ and $\alpha_{\mathfrak{p}} \in R_{g_{\mathfrak{p}}}$ such that $(\alpha_p, D(g)) = f_{\mathfrak{q}}$ for all $\mathfrak{q} \in D(g)$.

We define $\mathcal{O}_{\operatorname{Spec} R}$ to be the *structure sheaf* of $\operatorname{Spec} R$. $\mathcal{O}_{\operatorname{Spec} R}$ is a *contravariant functor* from

$$\{\text{open sets in } \operatorname{Spec} R \text{ with arrows being inclusion}\} \to \textbf{Ring}.$$

That is, the restriction map $\operatorname{res}_{U,V} \colon \mathcal{O}_{\operatorname{Spec} R}(U) \to \mathcal{O}_{\operatorname{Spec} R}(V)$, given by $f \mapsto f|_V$, whenever $V \subset U$. Clearly, restricting from U to V to W where $W \subset V \subset U$ would yield the composition $\operatorname{res}_{V,W} \operatorname{res}_{U,V} = \operatorname{res}_{U,W}$.

An *affine scheme* is the data $(|\operatorname{Spec} R|, \mathcal{O}_{\operatorname{Spec} R})$.

Theorem 8.4. We have the following properties of the structure sheaf,

(i) $\mathcal{O}_{\operatorname{Spec} R}(R(f)) \cong R_f$,

(ii) Given an open cover $\{U_\alpha\}_{\alpha\in\mathcal{A}}$ of U and $g_\alpha \in \mathcal{O}_{\operatorname{Spec} R}(U_\alpha)$ such that

$$g_\alpha|_{U_\alpha\cap U_\beta} = g_\beta|_{U_\alpha\cap U_\beta},$$

there is a *unique* $g \in \mathcal{O}_{\operatorname{Spec} R}(U)$ such that $g|_{U_\alpha} = g_\alpha$.

Proof. For (i), any element of $\mathcal{O}_{\operatorname{Spec} R}(D(f))$ induces an open coverign of $D(f)$ with basic open sets and elements that agree on overlaps, so we are done. Specifically, you need

$$R \to \prod_{\mathfrak{p}\in\operatorname{Spec} R} R_{\mathfrak{p}}$$

to be an injection.

For (ii), the definition of $\mathcal{O}_{\operatorname{Spec} R}$ necessitates this to be true. □

As an example, $\mathcal{O}_{\mathbb{A}^2}(\mathbb{A}^2 - \{(0,0)\})$. Then $\mathbb{A}^2_{\mathbb{C}} = \operatorname{Spec}\mathbb{C}[x,y]$. Here, $D(x)\cup D(y) = \mathbb{A}^2\setminus\{(0,0)\}$, for which R is given by $f \in C[x,y]_x$ and $g \in C[x,y]_y$ such that f, g agree on $D(xy)$, i.e., in $\mathbb{C}[x,y]_{xy}$. We need f, g as our functions.

If you know the answers to all the optional exercises, you know how to get a 100 on the midterm.

Day 9: Sheaves, (Locally) Ringed Spaces (Oct. 6, 2026)

This week is the last week of material before the midterm. The midterm will consist of exercises polished into a document (some are too trivial, some are unclear, etc.). So!![5]

We are now going to discuss the structure sheaf $\mathcal{O}_X$ on the spectrum $X = \operatorname{Spec} A$.

Definition 9.1. Given a topological space X, a *presheaf* is a contravariant functor $\mathscr{F} : \mathbf{Open}(X) \to \mathbf{Ab}$, where $\mathbf{Open}(X)$ denotes the open sets of X and $\mathbf{Ab}$ denotes abelian groups, with morphisms being inclusions. Given $V \subset U$, we write $f \mapsto f|_V$ for the map $\mathscr{F}(U) \to \mathscr{F}(V)$.

Let us give some examples of presheaves.

(i) Given an abelian group A, we have that

$$A : \mathbf{Open}(X) \to \mathbf{Ab}, \quad U \mapsto A,$$

where restriction is identity.

(ii) The bounded continuous functions on a manifold forms a presheaf as well. Restrictions of bounded continuous functions are bounded continuous functions, after all.

(iii) Any assignment of open sets to abelian groups, $\mathbf{Open}(X) \to \mathbf{Ab}$, where the restrictions are 0.

"Presheaves have nothing to do with functions. They're just some random way to assign abelian groups to functions such that restriction behaves."

A morphism of presheaves $\mathscr{F} \to \mathscr{G}$ is a *natural functor* where, for each U, we assign $\mathscr{F}(U) \to \mathscr{G}(U)$ such that there is compatibility as follows,

$$\begin{array}{ccc} \mathscr{F}(U) & \longrightarrow & \mathscr{G}(U) \\ \downarrow & & \downarrow \\ \mathscr{F}(V) & \longrightarrow & \mathscr{G}(V) \end{array}$$

He then said something about $\mathbb{Z}_M$ and bounded continuous functions on M, but I hope that's not relevant.

Definition 9.2. A *sheaf* is a presheaf satisfying the unique gluing condition: let $(U_\alpha)_{\alpha \in \mathcal{A}}$ be a cover of U, and consider $f_\alpha \in \mathscr{F}(U_\alpha)$ satisfying

$$f_\alpha\big|_{U_\alpha \cap U_\beta} = f_\beta\big|_{U_\alpha \cap U_\beta}.$$

Then there exists some unique $f \in \mathscr{F}(U)$ such that $f|_{U_\alpha} = f_\alpha$ for all α.

We now want to check if the following presheaves are sheaves.

(i) Consider $X = \mathbb{R}$ and $A = \mathbb{Z}$. Consider the empty covering of the empty set; then the gluing condition is vacuous, meaning that any $f \in \mathscr{F}(U)$ would satisfy $f|_{U_\alpha} = f_\alpha$ for all α, for which there are no U_α at all, since the cover is empty. Thus, it is not a sheaf. He said this was a bad example.

(ii) Let's check the bounded continuous functions on a manifold. Let $X = \mathbb{R}$, and let $U_i = (i, i+2)$ be a bunch of (arbitrarily chosen) intervals that cover $\mathbb{R}$. We can just consider $f_i = x$ on each of the intervals; clearly, the f_i agree on the overlaps and are bounded on their respective intervals, but this obviously does not glue to a bounded continuous function (if it did glue to a function, it would have to glue to x, which is *not* a bounded continuous function).

[5] and he did a fruity-ass jump (i mean a hop)

(iii) Producing compatible gluing data in the last case is super easy (i.e., in the case of assigning open sets to abelian groups); but, we will have a "massive failure of uniqueness", i.e., any candidate $f \in \mathscr{F}(U)$ has $f|_{U_\alpha} = 0 \neq f_\alpha$. Thus, we see that this is generally not a sheaf (for example, if X has the indiscrete topology, then it has to be a sheaf; to quote Hunter, "there are always ways to make terrible sheaves").

We now want to talk about the abelian group of germs. The germs at $x \in X$ are pairs (f, U) where $x \in U$ and $f \in \mathscr{F}(U)$. $(f, U) \sim (g, V)$ if there exists $W \subset U \cap V$ such that $f|_W = g|_W$. $\mathscr{F}_p$ is then the abelian group of germs (it is a ring if $\mathscr{F}$ is valued in rings). We now talk about sheafification.

Given a presheaf $\mathscr{F}$, we define a sheaf $\mathscr{F}^\#$ given by the compatible collections of germs. Specifically, for $(f_x)_{x\in U}$, $f_x \in \mathscr{F}_U$, such that, for all x, there exists $V \subset U$, $g \in \mathscr{F}(V)$ such that $(g, V) = f_x$ for all $x \in V$.

A *ringed space* is a pair $(X, \mathcal{O}_X)$, where $\mathcal{O}_X$ is the sheaf of rings. As an example, consider $(M, C^2(M, \mathbb{R}))$ or $(\operatorname{Spec} R, \mathcal{O}_{\operatorname{Spec} R})$. The problem is that the vanishing locus of closed sets need not be closed, since we've defined two pieces of incompatible data. We need to impose an extra condition.

Definition 9.3. A *locally ringed space* is a ringed space where the germs at x, $\mathcal{O}_x$, are a local ring for all $x \in X$. Recall that a local ring is a ring with a unique maximal ideal $\mathfrak{m}$.

We're going to use the localness of this ring to define what it means to vanish or not vanish at a point. We may define $\kappa(x)$, in the case that X is a local ring, to be $\mathcal{O}_x/\mathfrak{m}$.

Note that $I \subset R$ is the unique maximal ideal of R if and only if all elements of $R \setminus I$ are units. For example, $\mathcal{O}_{M,p}$, where M is a manifold and $p \in M$ is a local ring with maximal ideal given by $\{(f, U) \mid f(p) = 0\}$, i.e., germs (f, U). It is a local ring because if we have (f, U) with $f(p) \neq 0$, then continuity implies that $f \neq 0$ on a neighborhood $V \subset U$ of p; with this, we may then write that the inverse of (f, V) is (f^{-1}, V).

Hence, $\mathbb{Z}^\#$ need not be a locally ringed space, as inverses don't exist locally.

Theorem 9.4. If $f \in \mathcal{O}_X(X)$ and X is locally ringed, then $D(f) \subset X$ is open.

Proof. If $f(p) \neq 0$ in $\kappa(x)$, i.e., $p \in D(f)$, then f is a unit in $\mathcal{O}_{X,p}$. (f, U) is a unit implies that there is a potentially smaller $V \subset U$, such that the germ (f^{-1}, V) on V has $f \neq 0$, since $ff^{-1} = 1$. □

Day 10: Locally Ringed Maps (Oct. 8, 2026)

Today, we're going to describe how to "do morphisms" between affine schemes. Then we're going to get to schemes!

Let $(X, \mathcal{O}_X)$ be a locally ringed space, where X is a topological space and $\mathcal{O}_X$ is the associated sheaf of rings. The germs at x, $\mathcal{O}_{X,x}$, is a local ring. $D(f) = \{x \in X \mid f(x) \neq 0\}$ is open, and $f(x)$ has its image in $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}$.

Definition 10.1. Given a presheaf $\mathscr{F}$ on X and a continuous map $f\colon X \to Y$, $f_*\mathscr{F}$ is the presheaf given by $(f_*\mathscr{F})(V) \coloneqq \mathscr{F}(f^{-1}(V))$.

Theorem 10.2. If $\mathscr{F}$ is a presheaf, then $f_*\mathscr{F}$ is a presheaf. If $\mathscr{F}$ is a sheaf, then $f_*\mathscr{F}$ is a sheaf.

Proof. Given $W \subset V \subset U$, we want to show that, for $\alpha \in f_*\mathscr{F}(U)$,

$$\left(\alpha_U\big|_V\right)\big|_W = \alpha\big|_W, \quad \alpha_U \in \mathscr{F}(f^{-1}(U)).$$

$\mathscr{F}(f^{-1}(U)) = f_*\mathscr{F}(U)$ restricts down to $\alpha_U|_V \in \mathscr{F}(f^{-1}(V)) = f_*\mathscr{F}(V)$, which restricts down to $\left(\alpha_U|_V\right)\big|_W \in \mathscr{F}(f^{-1}(W)) = f_*\mathscr{F}(W)$. By the presheaf property for $\mathscr{F}$, we get what we want.

To check the implication for sheaves, let $\{U_\alpha\}_{\alpha\in\mathcal{A}}$ be a covering with $U \subset Y$. We want to show that $g_\alpha \in (f_*\mathscr{F})(U_\alpha)$ such that $g_\alpha|_{U_\alpha\cap U_\beta} = g_\beta\big|_{U_\alpha\cap U_\beta}$, so there exists a unique g such that $g|_{U_\alpha} = g_\alpha$. Thus, $g_\alpha \in \mathscr{F}(f^{-1}(U_\alpha))$ satisfies

$$g_\alpha\big|_{f^{-1}(U_\alpha\cap U_\beta)} = g_\beta\big|_{f^{-1}(U_\alpha\cap U_\beta)},$$

for which $f^{-1}(U_\alpha \cap U_\beta) = f^{-1}(U_\alpha) \cap f^{-1}(U_\beta)$. It follows that $f^{-1}(U) = \bigcup f^{-1}(U_\alpha)$, on which we have compatible guing data, so there exists a unique $g \in \mathscr{F}(f^{-1}(U)) = f_*\mathscr{F}(U)$ restricting correctly. □

Definition 10.3. A map of ringed spaces $(X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is the data of continuous $f\colon X \to Y$ and a morphism of sheaves $\mathcal{O}_Y \to f_*\mathcal{O}_X$. Specifically, they are compatible ring homomorphisms $\mathcal{O}_Y(U) \to \mathcal{O}_X(f^{-1}(U))$.

A map of locally ringed spaces $(X, \mathcal{O}_x) \to (Y, \mathcal{O}_Y)$ is a map of ringed spaces such that the induced map from $f_*^\#\colon \mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$ is a local map of local rings, meaning the preimage of a maximal ideal under $f_*^\#$ is a maximal ideal as well.

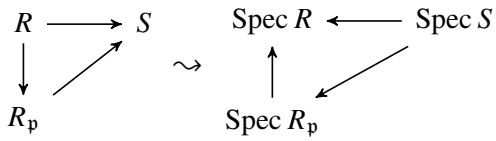
The map $\mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$ takes a germ (g, U) (where $f(x) \in U$), and under the map $\mathcal{O}_Y(U) \to \mathcal{O}_X(f^{-1}(U))$, g maps to $g' \in \mathcal{O}_X(f^{-1}(U))$. Treating this as a germ, we obtain a germ $(g', f^{-1}(U))$ at $x \in X$, which gives the element of $\mathcal{O}_{X,x}$.

We now have a notion of pulling back germs via the above procedure. Now, consider the pullback map on germs of functions $f_x^\#\colon \mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$; $\mathfrak{m}_x$ denotes the germs vanishing at x and $\mathfrak{m}_{f(x)}$ denotes the germs vanishing at $f(x)$, in X and Y respectively. If $(f_{f(x)}^\#)^{-1}(\mathfrak{m}_x) \subsetneq \mathfrak{m}_{f(x)}$, then there exists $\alpha \in \mathfrak{m}_{f(x)}$ such that $f_{f(x)}^\#(\alpha) \notin \mathfrak{m}_x$. Thus, $f_x^\#$ being local means germs vanishing at $f(x)$ pullback to germs vanishing at x. If a germ doesn't vanish at $f(x)$, though, then its a unit, and so up to a unit, it does not vanish. Essentially, the vanishing and non-vanishing of germs is preserved under the pullback.

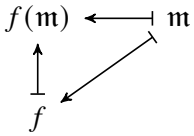
Theorem 10.4. Let $(X, \mathcal{O}_X)$ be a locally ringed space. A map $(X, \mathcal{O}_X) \to (\operatorname{Spec} R, \mathcal{O}_{\operatorname{Spec} R})$ is entirely specified by $R \to \mathcal{O}_X(X)$ and vice versa.

Proof. We claim that, if $(S, \mathfrak{m})$ is a local ring, and $f\colon R \to S$ is a ring map, then $f(\mathfrak{m}) \in \operatorname{Spec} R$ is the unique prime such that there is an induced local ring map $R_{f(\mathfrak{m})} \to S$.

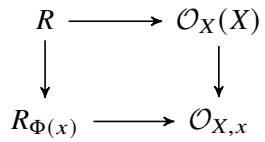
To prove this, we're going to take the spectrum and "see that this is really really obvious". Consider the map $\operatorname{Spec} S \to \operatorname{Spec} R$ (from the ring map f).



i.e., the diagram on the left induces that on the right by taking the spectrum (and that we take $\operatorname{Spec} R_{\mathfrak{p}}$ to be the primes contained in $\mathfrak{p}$). The map $\operatorname{Spec} S \to \operatorname{Spec} R_{\mathfrak{p}}$ sends $\mathfrak{m} \in \operatorname{Spec} S$ to $\mathfrak{p}$, since $\operatorname{Spec} S \to \operatorname{Spec} R$ must map $\mathfrak{m}$ to something contained in $\mathfrak{p}$. It follows that we have



Given $f\colon R \to \mathcal{O}_X(X)$, we want to (i) construct a map on topological spaces $\operatorname{Spec} R \to X$, (ii) show that it is continuous, and (iii) show that it is a map of locally ringed spaces. For (i), we can draw the following diagram,



If we look at $\Phi(x) = f^{-1}(\text{maximal ideal of } \mathcal{O}_{X,x})$, then $R_{\Phi(x)} \to \mathcal{O}_{X,x}$ is a local map of local rings. For (ii), we want to check that this "random assignment of Φ is continuous". Specifically, we want to show that $\Phi^{-1}(D(f))$ is open. Indeed, we showed that $\Phi^{-1}(D(g)) = D(f(g))$, and that $x \in \Phi^{-1}(D(g))$ is equivalent to $\Phi(x) \in D(g)$, which itself is equivalent to $g(\Phi(x)) \neq 0$. By the diagram (and the localness of the bottom map), we see that if g does not vanish in $R_{\Phi(x)}$, then it doesn't vanish at $\mathcal{O}_{X,x}$. Specifically, $f(g)(x)$ not vanishing at x is equivalent to $D(f(g))$.

Finally, for (iii), we construct $\mathcal{O}_{\operatorname{Spec} R} \to f_*\mathcal{O}_X$. Given $U \subset X$, we have $\mathcal{O}_{\operatorname{Spec} R}(U) \to \mathcal{O}_X(\Phi(U))$. This means the compatible germs on U are mapped from $(g_p)_{p\in U} \mapsto \Phi^{-1}(u)$. Given $x \in \Phi^{-1}(u)$, i.e., $\Phi(x) \in u$, take the image of $g_{\Phi(x)}$ in $\mathcal{O}_{X,x}$. If $(r/h^i, U)$ is a representative for a germ around p such that $(r/h^i, U) = g_q$ for all $q \in U$, then

$$\left(\frac{f(r)}{f(h^i)}, D(f(h))\right)$$

is the germ verifying compatibility on the image. □

"I'll say this was a confusing proof. I'll write this up nicely in the notes... my blood sugar dropped to half of what a normal human should have, but... as penance, I'll write this up nicely."

Theorem 10.5. Spec is a fully faithful functor from **Ring** to locally ringed spaces.

Definition 10.6. A *scheme* is a locally ringed space, which has an open covering by $\operatorname{Spec} R$ for various R.

Corollary 10.7. $\operatorname{Spec} \mathbb{Z}$ is the terminal object in the category of locally ringed spaces. This means there is always a unique map from $X \to \operatorname{Spec} \mathbb{Z}$.

Day 11: Schemes and Projective Space (Oct. 13, 2026)

Let us consider a map of locally ringed spaces, $(X, \mathcal{O}_X) \to (\operatorname{Spec} R, \mathcal{O}_{\operatorname{Spec} R})$. This is equivalent to a map $R \to \mathcal{O}_X(X)$. We will talk extensively about the global sections of $\mathcal{O}_X$, denoted $\Gamma(\mathcal{O}_X)$. In particular, $\Gamma(\mathscr{F})\colon \mathscr{F}(X)$.

Spec is a fully faithful functor from **Rings** $\to$ **LRS**, where **LRS** is the category of locally ringed spaces.

Definition 11.1. A *scheme* $(X, \mathcal{O}_X)$ is a locally ringed space such that there exists an open cover by $\{U_\alpha\}_{\alpha \in \mathcal{A}}$, and there are isomorphisms $(U_\alpha, \mathcal{O}_{U_\alpha}) \cong (\operatorname{Spec} R_\alpha, \mathcal{O}_{\operatorname{Spec} R_\alpha})$, for some rings R_α.

Note that $\mathcal{O}_{U_\alpha} = \mathcal{O}_X|_{U_\alpha}$. In particular, the image of $\Gamma(\mathscr{F})$ is an *affine scheme*, and we can think of a scheme as "gluing together a bunch of affine schemes". Let us look at some examples.

(i) Consider the affine space $\mathbf{A}^n_R \coloneqq \operatorname{Spec} R[x_1, \ldots, x_n]$.

(ii) Consider $D(x) = \mathbf{A}^1_{\mathbb{Z}} \setminus \{(x)\}$, where $\mathbf{A}^1_{\mathbb{Z}} = \operatorname{Spec} \mathbb{Z}[x]$. We may glue $\mathbf{A}^1_{\mathbb{Z}}$ to $\mathbf{A}^1_{\mathbb{Z}}$ along $D(x)$ to get the line with two origins (since $D(x)$ is simply excising the origin, and we identify the second copy of $\mathbf{A}^1_{\mathbb{Z}}$ with itself everywhere else). We can check that this is *not* an affine scheme: consider an open cover by the two copies of $\mathbf{A}^1_{\mathbb{Z}}$. Then the global sections are given by

$$\Gamma\mathcal{O}_{\text{doubled line}} = \left\{(f, g) \in \mathbb{Z}[x]^2 \mid f\big|_{D(x)} = g\big|_{D(x)}\right\}.$$

Since $\mathbb{Z}[x]$ is an integral domain and $D(x)$ is a dense open set, we see that $f = g$.
Thus, $\Gamma\mathcal{O}_{\text{doubled line}} = \mathbb{Z}[x]$. Moreover, we get a map

$$\begin{array}{ccc} \text{Doubled line} & \longrightarrow & (\operatorname{Spec} \mathbb{Z}[x], \mathcal{O}_{\operatorname{Spec} \mathbb{Z}[x]}) \\ \cup & \overset{\sim}{\nearrow} & \\ \mathbf{A}^1_{\mathbb{Z}} & & \end{array}$$

Note that the map from the doubled line to $(\operatorname{Spec} \mathbb{Z}[x], \mathcal{O}_{\operatorname{Spec} \mathbb{Z}[x]})$ cannot be an isomorphism since the doubled line has an extra point.

(iii) Glue $\mathbf{A}^1_R$ along $\mathbf{A}^1_R$,respectively thought of as $\operatorname{Spec} R[t]$ and $\operatorname{Spec} R[s]$, along $\mathbf{A}^1_R$ minus its origin. By observing that $\operatorname{Spec} R[t, t^{-1}] \subset \operatorname{Spec} R[t]$ and $\operatorname{Spec} R[s, s^{-1}] \subset \operatorname{Spec} R[s]$, we see we may glue by sending $t \mapsto s^{-1}$ at the level of $\operatorname{Spec} R[t, t^{-1}] \to \operatorname{Spec} R[s, s^{-1}]$. As a result, we obtain the projective line $\mathbb{P}^1_R$.

We now compute the global sections. By definition,

$$\Gamma\mathcal{O}_{\mathbb{P}^1_R} = \left\{(f(s), g(t)) \mid f(s) = g(s^{-1})\right\}.$$

Concretely, we can visualize this as taking $g|_{D(t)}$, and mapping it to $f(s)|_{D(s)}$. Checking this coefficient by coefficient, we see that f only consists of nonnegative powers of s, and g only consists of nonpositive powers of s, where we must have that $f = g \in R$. Thus, $\Gamma\mathcal{O}_{\mathbb{P}^1_R} = R$.

Note that it is extremely important to understand the last example about the projective line (and spaces henceforth).

Theorem 11.2. Given locally ringed spaces $\{X_\alpha\}_{\alpha\in\mathcal{A}}$, open subsets $X_{\alpha\beta}\subset X_\alpha$, where $X_{\alpha\alpha}\coloneqq X_\alpha$ and $X_{\alpha\beta}$ "acts like" $X_\alpha\cap X_\beta\subset X_\alpha$, and isomorphisms $\varphi_{\alpha\beta}\colon X_{\alpha beta}\to X_{\beta\alpha}$ satisfying

$$\begin{array}{ccc} X_{\alpha\beta}\cap X_{\alpha\gamma} & \xrightarrow{\varphi_{\alpha\gamma}} & X_{\gamma\alpha}\cap X_{\gamma\beta} \\ & {\scriptstyle\varphi_{\alpha\beta}}\searrow \quad \swarrow{\scriptstyle\varphi_{\beta\gamma}} & \\ & X_{\beta\alpha}\cap X_{\beta\gamma} & \end{array}$$

then there exists a unique locally ringed space X with open cover $\{U_\alpha\}_{\alpha\in\mathcal{A}}$ and isomorphisms $\psi_\alpha\colon U_\alpha\to X_\alpha$ such that $\psi_\alpha(U_\alpha\cap U_\beta)=X_{\alpha\beta}$ and $\varphi_{\alpha\beta}=\psi_\beta\big|_{U_\alpha\cap U_\beta}\psi_\alpha^{-1}\big|_{X_{\alpha,\beta}}$.

This thing is *somewhere* on the stacks project.

Definition 11.3 (Projective n-space)**.** Consider $S\coloneqq R[x_1^{\pm1},\dots,x_n^{\pm1}]_{(0)}\subset R[x_1^{\pm1},\dots,x_n^{\pm1}]$. Note that this is a degree 0 subring if $\deg R=0$. Note that $\deg x_i=1$ and $\deg x_i^{-1}=-1$. To get the projective n-space, we're going to try to construct a map $(\mathbf{A}^1\setminus\{0\})^n\to\mathbf{A}^{n-1}$.

Denote R_i to be the subring of S where x_i is allowed to be a denominator, R_{ij} to be the subring where x_ix_j is allowed to be a denominator, and so on for any collection of indices in the subscript. Then

$$R_i=R\left[\frac{x_0}{x_i},\dots,\widehat{\frac{x_i}{x_i}},\dots,\frac{x_n}{x_1}\right].$$

Observe that $R_i[(x_j/x_i)^{-1}]=R_{ij}=R_j[(x_i/x_j)^{-1}]$. This means we may "trade off" arbitrary powers of x_i and x_j to get Laurent polynomials in x_i and x_j to construct R_{ij}. Also observe that

$$D\left(\frac{x_j}{x_i}\right)_{\operatorname{Spec}R_i}=\operatorname{Spec}R_{ij}=D\left(\frac{x_i}{x_j}\right)_{\operatorname{Spec}R_j}.$$

We define φ_{ij} to be this map on spectra, corresponding to the string of equalities above. The gluing condition would then be

$$R_i\left[\left(\frac{x_j}{x_i}\right)^{-1}\right]\left[\left(\frac{x_k}{x_i}\right)^{-1}\right]=R_j\left[\left(\frac{x_i}{x_j}\right)^{-1}\right]\left[\left(\frac{x_k}{x_j}\right)^{-1}\right]=R_k\left[\left(\frac{x_i}{x_k}\right)^{-1}\right]\left[\left(\frac{x_j}{x_k}\right)^{-1}\right],$$

where all three of these are equal to R_{ijk}.

Theorem 11.4. $\Gamma\mathcal{O}_{\mathbb{P}^n_R}=R$.

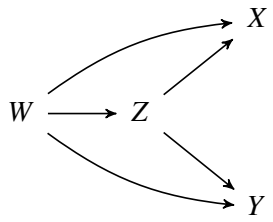
Proof. On an open covering of $\operatorname{Spec}R$, we see that the $f_i\in R_i$, which are compatible in $\operatorname{Spec}R_{ij}=\operatorname{Spec}R_i\cap\operatorname{Spec}R_j\subset\mathbb{P}^n_R$ have

$$f_i\big|_{\operatorname{Spec}R_{ij}}=f_j\big|_{\operatorname{Spec}R_{ij}},$$

as $R_i,R_j\hookrightarrow R_{ij}$. Thus, $f_i=f_j$, and so $\Gamma\mathcal{O}_{\mathbb{P}^n_R}=\bigcap_i R_i=R$. □

Day 12: Fiber Products (Oct. 15, 2026)

Today, we will talk about products of schemes. Let C be a category with objects X and Y. We define a product Z, if it exists, to be an object for X and Y, equipped with maps $Z \to X$ and $Z \to Y$ such that if $W \to X$ and $W \to Y$ are 2 maps in C, then there exists a unique map $W \to Z$ such that



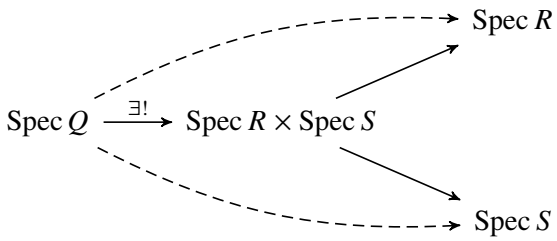
commutes. As an example, given sets S, T, there is a product $S \times T = \{(s,t) \mid s \in S, t \in T\}$, along with projections π_S, π_T onto S, T respectively, such that the above diagram makes sense, i.e., $W \to S \times T$ (where $S \times T$ was originally Z) is given by $w \mapsto (f(w), g(w))$.

Let us look at the category of affine schemes, **AffSch** = **Rings**^{op}.

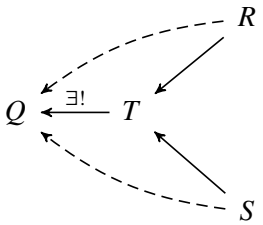
Theorem 12.1. Products are unique up to unique isomorphisms if they exist.

The proof is given by symbol pushing through diagrams. I'm not going to draw diagrams for this.

From the above, we see that if a product exists, we may call it $X \times Y$. Now, consider



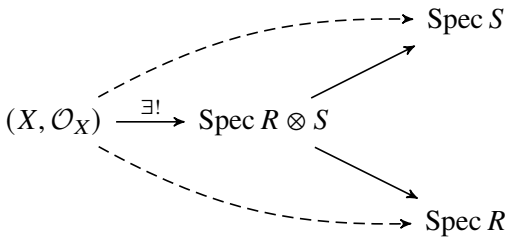
where the unique map $\text{Spec}\, Q \to \text{Spec}\, R \times \text{Spec}\, S$ induces the commutativity in the diagram above. Denoting $\text{Spec}\, R \times \text{Spec}\, S$ as $\text{Spec}\, T$, we see that

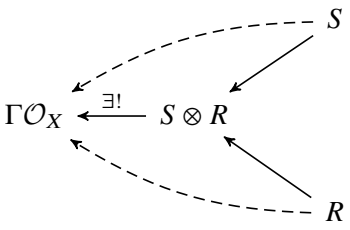


where $T = R \otimes S$. Specifically, the tensor product behaves as usual, i.e., $R \to R \otimes S$ sends $r \mapsto r \otimes 1$, and $S \to R \otimes S$ sends $s \mapsto 1 \otimes s$.

Theorem 12.2. $\operatorname{Spec}(R \otimes S) = \operatorname{Spec} R \times \operatorname{Spec} S$ in the category of schemes.

Proof. We prove this by drawing diagrams.





□

In this manner, $\mathbf{A}^1_{\mathbb{Z}} \times \mathbf{A}^1_{\mathbb{Z}} \cong \mathbf{A}^2_{\mathbb{Z}}$, i.e., $\operatorname{Spec}\mathbb{Z}[x] \times \operatorname{Spec}\mathbb{Z}[y] = \operatorname{Spec}\mathbb{Z}[x, y]$. In terms of topology, though this is not a product topological space (i.e., with a basis of closed sets $Z_1 \times Z_2$). Instead, we have a map of topological spaces from $\mathbf{A}^2_{\mathbb{Z}}$ to each of $\mathbf{A}^1_{\mathbb{Z}}$, so we have a topological product of $\mathbf{A}^1_{\mathbb{Z}}$ and $\mathbf{A}^1_{\mathbb{Z}}$.

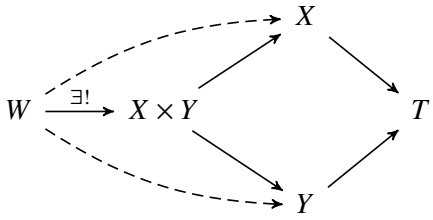
The important part is like, the product of these topological spaces in terms of point × point, line × line, point × line, and so on; they don't reach some of the primes in $\mathbb{Z}[x, y]$. However confusingly this is worded.

As an exaxmple, let us consider the product of spectra $\operatorname{Spec}\mathbb{F}_2 \times \operatorname{Spec}\mathbb{F}_3$. Since $\mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z}/3\mathbb{Z} = \mathbb{Z}/((2)+(3)) = 0$, we get that $\operatorname{Spec}\mathbb{F}_2 \times \operatorname{Spec}\mathbb{F}_3 = \varnothing$, i.e., the spectrum of the zero ring.

Now, consider $\operatorname{Spec} R \times \operatorname{Spec}\mathbb{Z} = \operatorname{Spec} R \otimes \mathbb{Z} = \operatorname{Spec} R$. This comes from the fact that $\operatorname{Spec}\mathbb{Z}$ is the terminal object in the category.

For a last example, let us look at $\mathbf{A}^1_{\mathbb{C}} \times \mathbf{A}^1_{\mathbb{C}}$. This is $\operatorname{Spec}\mathbb{C}[x] \otimes_{\mathbb{Z}} \mathbb{C}[y]$... which is... "impossibly bad to describe". Instead, we want $\operatorname{Spec}\mathbb{C}[x] \otimes_{\mathbb{C}} \mathbb{C}[y]$, which is $\mathbf{A}^2_{\mathbb{C}}$.

We now introduce the fiber product. $X \times_T Y$, if it exists, is the unique product with maps $X \times_T Y$ to X and Y such that



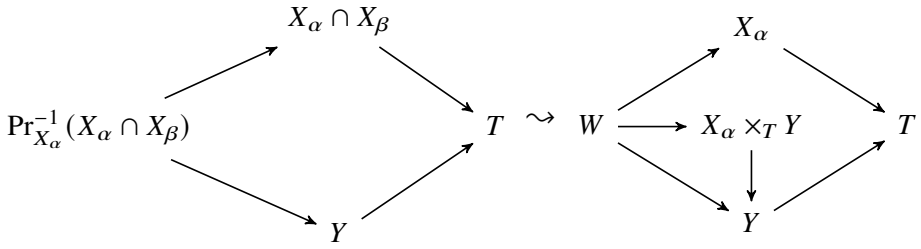
For sets $X \times_T Y = \{(x, y) \mid h_X(x) = h_Y(y)\}$, where h_X, h_Y are the maps from $X \to T$ and $Y \to T$ respectively, we have that the above diagram with arrows flipped around for spectra. (which I don't want to draw, see p.87 Hartshorne)

As a quick note, if $E \in \mathscr{C}$ is terminal, then $C \times_E D \cong C \times D$.

Theorem 12.3. Fiber products $X \times_T Y$ exist in schemes.

Proof. If X, Y, T are affine, we are done. If X is not affine but Y, T are, then take an open cover $\{X_\alpha\}_{\alpha \in \mathcal{A}}$ of X. Then $X_\alpha \times_T Y$ exists for each α. We claim that $(X_\alpha \cap X_\beta) \times_T Y$ also exists.

Indeed,



The image lands in the open set $\mathrm{Pr}_{X_\alpha}^{-1}(X_\alpha \cap X_\beta)$, since $W \to X_\alpha \times_T Y \to X_\alpha$ has image in $X_\alpha \cap X_\beta$ by the above diagrams. From here, we may do "some horrible check" to see that we are done.

In that next case, if X is not affine, T is affine, and Y is not, then Y has an open cover, and we may run the same argument as before. Finally, if T is *also* not affine, then cover T with $\{T_\alpha\}_{\alpha \in \mathcal{A}}$. Then X_α is the preimage of T_α in X, and Y_α is the preimage of T_α in Y. Then $X_\alpha \times_{T_\alpha} Y_\alpha = X_\alpha \times_T Y$. □

Morally, if you want to cover $X \times_T Y$ with open affines, then you first cover T with open affines. Given an open $\operatorname{Spec} R \subset T$, cover preimages of X and Y with open affines $\operatorname{Spec} A_i$ and $\operatorname{Spec} B_j$, then you get an open cover $\operatorname{Spec} A_i \otimes_R B_j$.

As a corollary of the construction, $\mathbb{P}^n_R = \operatorname{Spec} R \times_{\operatorname{Spec} R} \mathbb{P}^n_{\mathbb{Z}}$. $\mathbb{P}^n_R$ can be covered with

$$\operatorname{Spec} R \otimes_{\mathbb{Z}} \mathbb{Z}\left[\frac{x_0}{x_i}, \ldots, \widehat{\frac{x_i}{x_i}}, \ldots, \frac{x_n}{x_i}\right] = R\left[\frac{x_0}{x_i}, \ldots, \widehat{\frac{x_i}{x_i}}, \ldots, \frac{x_n}{x_i}\right].$$

In this way, we may define $\mathbb{P}^N_X \coloneqq X \times \mathbb{P}^n_{\mathbb{Z}}$.

Day 13: Graded Rings and Group Schemes (Oct. 20, 2026)

Today, we're going to be talking about the Proj construction. This generalizes the construction of projective space by gluing together the open sets we saw last week.

What does it mean to be a graded ($\mathbb{N}$, if no qualifications are given), or $\mathbb{Z}$-graded ring? What is the geometric interpretation? The action of "multiplication by ψ_m on your space"...; classically,

$$\mathbb{P}^n_{\mathbb{C}} \cong (\mathbb{C}^{n+1} \setminus \{0\})/\mathbb{C}^*,$$

which can be read off as homogeneous coordinates of the form $[x_0 : x_1 : \cdots : x_n]$, with $(x_0, \ldots, x_n)$ not all zero. We may cover $\mathbb{P}^n_{\mathbb{C}}$ by $\{x_i \neq 0\}$, defined as

$$D_+(x_i) = \left[\frac{x_0}{x_i} : \cdots : 1 : \cdots : \frac{x_n}{x_i}\right] \cong \mathbf{A}^n_{\mathbb{C}}.$$

On $\mathbb{P}^n_{\mathbb{C}}$, the zero locus of the homogeneous polynomials makes sense. For example,

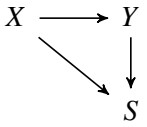
$$Z_{+,\mathbb{C}}\left(x^3 + x_1 x = x_3\right),$$

where the subscripts mean "considered in projective space", makes sense. Now, the action of $\mathbb{C}^*$ on $\mathbf{A}^{n+1}_{\mathbb{C}}$ is equivalent to giving a grading to $\mathbb{C}[x_0, \ldots, x_n]$ so that the traded piece consists of homogeneous polynomials of degree i, whose zero locus makes sense on all of $\mathbb{P}^n_{\mathbb{C}}$. Specifically,

$$f(\lambda \cdot x) = \sum \lambda^i f_i(x),$$

where $\lambda \in \mathbb{C}^*$, $\cdot$ is the action on $\mathbf{A}^{n+1}_{\mathbb{C}}$, and the assignment "=" is unique. Note that $f_i(x)$ are also the homogeneous polynomials of degree 1.

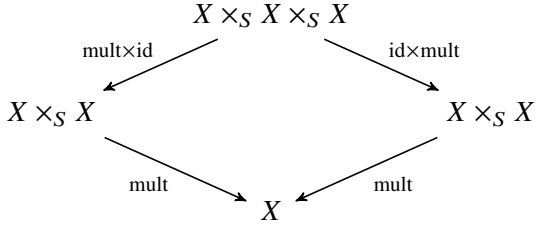
Now, consider the schemes over S, denoted $\mathbf{Sch}/S$, where S is a chosen scheme. The objects of this category are denoted X/S, which are the maps $X \to S$, where X are schemes. The morphisms are given by $X/S \to Y/S$, are given by connecting the diagram



Theorem 13.1. The fiber product $X \times_S Y$ is just the product $X/S \times Y/S$ in $\mathbf{Sch}/S$.

In this way, we may consider S to be the terminal object of $\mathbf{Sch}/S$.

Definition 13.2. A *group scheme* $X/S \in \mathbf{Sch}/S$ is an element of $\mathbf{Sch}/S$ equipped, equipped with morphisms $\text{mult}\colon X \times_S X \to X$, $\text{inv}\colon X \to X$, and $e\colon S/S \to X/S$ (as $S \to X \to S$ is the identity), and satisfies all the group axioms, i.e., the diagram commutes,



e, being the identity, would then have to satisfy

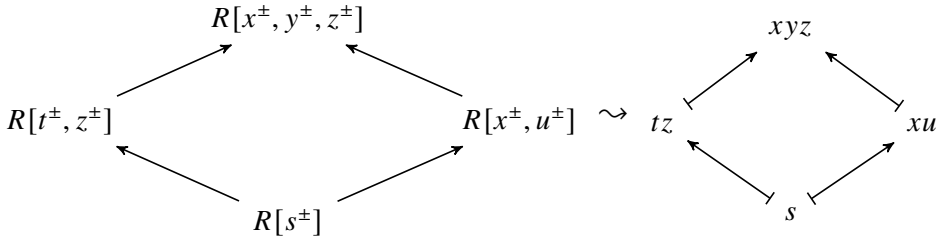
$$\begin{array}{ccccccc} X & \xrightarrow{\sim} & S \times_S X & \xrightarrow{e\times_S \mathrm{id}} & X \times_S X & \xrightarrow{\text{mult}} & X \\ & \searrow^{\sim} & & & & & \\ & & X \times_S S & \xrightarrow{\mathrm{id}\times_S e} & X \times_S X & \xrightarrow{\text{mult}} & X \end{array}$$

From here, we would have $G \to \{\phi\} \times G \mapsto G \times G \to G$, where $G \times G \to \{\phi\} \times G$ by φ. This means we would have the diagram

$$\begin{array}{ccccccc} X & \xrightarrow{\mathrm{id}\times\mathrm{id}} & X \times_S X & \xrightarrow{\text{inv}\times\mathrm{id}} & X \times_S X & \longrightarrow & X \\ & & & \searrow^{\mathrm{id}\times_S \text{inv}} & & & \\ & & & & X \times_S X & \longrightarrow & X \end{array}$$

We are now done defining this group scheme.[6] In $\mathbf{Sch}/\operatorname{Spec} R$, we have a group scheme $\mathbb{G}_{m,R} \coloneqq D(x) \subset \mathbf{A}^1_{m,R}$, where $\mathbb{G}_{m,R} = \operatorname{Spec} R[x^2]$ and $\mathbf{A}^1_{m,R} = \operatorname{Spec} R[x]$.

We want that $\mathbb{G}_{m,R} \times_{\operatorname{Spec} R} \mathbb{G}_{m,R} \to \mathbb{G}_{m,R}$ as $R[x^\pm, y^\pm] \leftarrow R[t^\pm]$, where $xy \leftarrow\!\shortmid t$. We want this to satisfy the associativity diagram,



In this way, we have the following theorem,

Theorem 13.3. A graded ring $S = \bigoplus_{i\in\mathbb{Z}} S_i$ is the same as data $\mathbb{G}_{m,\operatorname{Spec}\mathbb{Z}} \curvearrowright \operatorname{Spec} S$. A $\mathbb{N}$-graded ring is equivalent to $\mathbb{G}_{k,\operatorname{Spec}\mathbb{Z}} \curvearrowright \operatorname{Spec} S$. This is equivalent to asking that the following diagram commutes,

$$\begin{array}{ccc} & & \mathbf{A}^1_{\operatorname{Spec} Z} \times \operatorname{Spec} S \\ & \overset{\exists!}{\nearrow} & \downarrow \\ \mathbb{G}_{m,\operatorname{Spec}\mathbb{Z}} \times \operatorname{Spec} S & \longrightarrow & \operatorname{Spec} S \end{array}$$

This also works with R-algebras and $\mathbb{G}_{m,\operatorname{Spec} R}$.

[6] jfc